

NAMLAMEIS

Cybersecurity Website

Complete Development Roadmap
Junior Developer Edition — v1.0

5 Phases Project Structure	2 Months Timeline	Solo Dev Team Size	Cybersecurity Domain
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1. Project Overview

This document is the complete development roadmap for NAMLAMEIS, a cybersecurity-focused website built by a junior developer. It covers everything from initial environment setup to deployment, content strategy, and ongoing maintenance. The goal is a professional, credible platform in the cybersecurity space that showcases skills, publishes technical content, and can evolve into a resource hub.

1.1 Vision Statement

Build a clean, performant, security-first website where NAMLAMEIS publishes cybersecurity research, tutorials, CTF write-ups, and tools — establishing a credible personal brand in the security community.

1.2 Core Goals

- Establish a professional online presence as a cybersecurity practitioner
- Publish educational content (tutorials, write-ups, security guides)
- Demonstrate secure coding practices in the website itself
- Build a portfolio of tools and projects
- Grow an audience through consistent quality content

1.3 Tech Stack Decision

Layer	Technology	Reason
Frontend Framework	Next.js 14 (App Router)	SSR + SSG, great performance, SEO friendly
Styling	Tailwind CSS	Rapid development, consistent design system
CMS / Content	MDX (local files)	Markdown + JSX, version-controlled posts
Authentication	NextAuth.js (if needed)	Secure auth for admin panel
Database	PostgreSQL + Prisma	Structured data, type-safe queries
Hosting	Vercel	Free tier, great Next.js integration
Domain	Namecheap / Cloudflare	Affordable, DNS + CDN via Cloudflare
Security Headers	next-secure-headers	CSP, HSTS, X-Frame options
CI/CD	GitHub Actions	Automated testing + deployment pipeline
Monitoring	Sentry + Vercel Analytics	Error tracking, performance monitoring

2. Roadmap Phases at a Glance

Phase 1	Foundation & Setup	🕒 Weeks 1–2
Phase 2	Core Website Build	🕒 Weeks 3–4
Phase 3	Security Hardening	🕒 Weeks 5
Phase 4	Content & SEO Growth	🕒 Weeks 6
Phase 5	Launch & Iteration	🕒 Weeks 7–8

3. Phase 1 — Foundation & Setup

Months 1–2 | Objective: Get your local environment, tools, and repository ready before writing a single line of production code.

3.1 Development Environment

- Install Node.js LTS (v20+) via nvm for version management
- Set up VS Code with essential extensions:
 - ESLint, Prettier, Tailwind CSS IntelliSense, GitLens
 - Error Lens, REST Client, GitHub Copilot (optional)
- Configure WSL2 (Windows) or use native terminal (Mac/Linux)
- Install Git and configure SSH keys for GitHub

3.2 Project Initialization

Task	Command / Action	Notes
Create Next.js app	<code>npx create-next-app@latest namlameis</code>	Use App Router, TypeScript, Tailwind
Init Git repo	<code>git init && git remote add origin ...</code>	Connect to GitHub immediately
Install core deps	<code>npm install prisma @prisma/client</code>	Database ORM setup
Setup ESLint/Prettier	<code>npm install -D eslint-config-prettier</code>	Consistent code style
Configure env vars	Create <code>.env.local</code> file	Never commit secrets to Git
Create <code>.gitignore</code>	Add <code>.env*</code> , <code>node_modules</code> , <code>.next</code>	Standard ignores

3.3 Repository Structure

Adopt a clean folder structure from day one to avoid tech debt:

Folder / File	Purpose
<code>/app</code>	Next.js App Router pages and layouts
<code>/app/blog</code>	Blog listing and individual post pages
<code>/app/tools</code>	Security tools showcases
<code>/components</code>	Reusable React components
<code>/content/posts</code>	MDX blog post files (version controlled)
<code>/lib</code>	Utility functions, API helpers
<code>/public</code>	Static assets (images, icons, robots.txt)
<code>/.github/workflows</code>	GitHub Actions CI/CD pipeline files
<code>/prisma</code>	Database schema and migrations

4. Phase 2 — Core Website Build

Months 3–5 | Objective: Build all visible pages, components, and the blog system. This is the main construction phase.

4.1 Page Architecture

Page	URL	Priority	Description
Home / Landing	/	P1	Hero, intro, latest posts, featured tools
About	/about	P1	Bio, skills, certifications, background
Blog Index	/blog	P1	List of all posts, categories, search
Blog Post	/blog/[slug]	P1	Individual MDX post with code highlighting
Tools	/tools	P2	Showcase of security scripts and utilities
Resources	/resources	P2	Curated links, learning paths
CTF Write-ups	/ctf	P2	Capture-the-flag challenge solutions
Contact	/contact	P3	Contact form with honeypot spam protection
404 Page	/404	P3	Custom branded not-found page

4.2 Component Library

- Layout Components: Navbar, Footer, Sidebar, MobileMenu
- Content Components: PostCard, PostMeta, TableOfContents, CodeBlock
- UI Components: Button, Badge, Alert, Tag, SearchBar
- Security Components: SecurityBadge, CVSSScore display, ThreatLevel
- SEO Components: Meta tags, Open Graph, JSON-LD schema

4.3 Blog / Content System

Use MDX for blog posts stored as files in /content/posts/. Each post has frontmatter:

Field	Type	Example
title	string	Introduction to SQL Injection
date	ISO date	2024-03-15
tags	array	["web", "injection", "owasp"]
category	string	"Web Security"
difficulty	enum	"beginner" "intermediate" "advanced"
excerpt	string	Short summary for cards and SEO
published	boolean	true / false (draft toggle)

5. Phase 3 — Security Hardening

Months 6–7 | Objective: As a cybersecurity website, security is a first-class feature. Harden the application and showcase your knowledge through your own security posture.

5.1 HTTP Security Headers

Header	Value / Config	Why It Matters
Content-Security-Policy	Strict allow-list of scripts/styles	Prevents XSS attacks
Strict-Transport-Security	max-age=31536000; includeSubDomains	Forces HTTPS everywhere
X-Frame-Options	DENY	Prevents clickjacking
X-Content-Type-Options	nosniff	Stops MIME sniffing attacks
Referrer-Policy	strict-origin-when-cross-origin	Controls referrer data leakage
Permissions-Policy	camera=(), microphone=(), geolocation=()	Disables unused browser APIs

5.2 Input Validation & Sanitization

- Validate all form inputs with Zod schema validation on server side
- Sanitize HTML content with DOMPurify before rendering
- Implement rate limiting on contact form (5 requests/min per IP)
- Add CSRF tokens to all state-changing requests
- Use honeypot fields to catch bot submissions

5.3 Dependency & Supply Chain Security

- Run npm audit weekly — fix critical and high vulnerabilities
- Enable Dependabot on GitHub for automated dependency PRs
- Pin dependency versions in package.json (avoid ^ ranges in prod)
- Use npm ci (not npm install) in CI/CD pipeline
- Review third-party scripts before adding (CSP enforcement)

5.4 Authentication Security (if admin panel added)

- Implement NextAuth.js with OAuth providers (GitHub / Google)
- Enforce MFA for admin account
- Set secure, HttpOnly, SameSite=Strict cookies
- Implement session timeout (30-minute inactivity)
- Log all admin login events with IP and user agent

5.5 Security Audit Checklist

Check	Tool / Method	Status
SSL/TLS configuration	SSL Labs scanner (ssllabs.com)	☐
HTTP headers audit	SecurityHeaders.com scan	☐
Dependency vulnerabilities	npm audit + Snyk	☐
OWASP Top 10 review	Manual code review checklist	☐
XSS vulnerability scan	OWASP ZAP passive scan	☐
DNS security (DNSSEC)	Cloudflare DNS settings	☐
Email security (SPF/DKIM)	MXToolbox check	☐
Privacy policy page	GDPR compliance review	☐

6. Phase 4 — Content & SEO Growth

Months 8–10 | Objective: Establish NAMLAMEIS as a reliable source of cybersecurity content. Grow organic traffic through SEO and consistent publishing.

6.1 Content Calendar — Topics to Cover

Category	Post Ideas	Cadence
Web Security	OWASP Top 10 series, XSS deep dive, SQLi lab	2x/month
CTF Write-ups	HackTheBox / TryHackMe solutions	After each solved box
Tools & Scripts	Python security tools, recon automation	1x/month
Career Advice	Getting started in cybersecurity, certifications	1x/month
News Analysis	CVE breakdowns, recent breach post-mortems	As needed
Tutorials	Burp Suite setup, Kali Linux tips, network scanning	1x/month

6.2 On-Page SEO Fundamentals

- Every post has unique title tag (50–60 chars) and meta description (150–160 chars)
- Implement structured data (JSON-LD) for articles and breadcrumbs
- Use semantic HTML: proper heading hierarchy (H1 > H2 > H3)
- Image optimization: WebP format, descriptive alt text, lazy loading
- Internal linking: each post links to 2–3 related posts
- Generate sitemap.xml automatically via next-sitemap package
- Submit sitemap to Google Search Console and Bing Webmaster Tools

6.3 Performance Targets

Metric	Target	Tool to Measure
Lighthouse Performance	≥ 90	Chrome DevTools / PageSpeed Insights
Largest Contentful Paint	< 2.5s	Core Web Vitals report
First Input Delay	< 100ms	Real User Monitoring
Cumulative Layout Shift	< 0.1	Lighthouse audit
Time to First Byte	< 600ms	GTmetrix
Bundle size (JS)	< 200KB gzipped	Next.js bundle analyzer

6.4 Community & Visibility

- Share posts on LinkedIn, Twitter/X, and r/netsec
- Engage in HackTheBox, TryHackMe, and CTF communities
- Add RSS feed (/rss.xml) for readers to subscribe
- Cross-post summaries to Dev.to or Medium with canonical link back

- Guest post on other cybersecurity blogs to build backlinks

7. Phase 5 — Launch & Iteration

Months 11–12 | Objective: Official public launch, gather feedback, iterate. Begin tracking metrics and planning next-stage features.

7.1 Pre-Launch Checklist

Area	Task	Done?
Performance	Run Lighthouse on all main pages, fix regressions	☐
Security	Run full security header scan + npm audit	☐
SEO	Verify sitemap, robots.txt, Google Search Console setup	☐
Cross-browser	Test on Chrome, Firefox, Safari, Edge	☐
Mobile	Test on iOS Safari and Android Chrome	☐
Accessibility	Run axe DevTools scan, fix critical a11y issues	☐
Content	Proofread all published posts, check broken links	☐
Analytics	Vercel Analytics or Plausible set up and verified	☐
Error monitoring	Sentry configured and first test error captured	☐
Backup	Database backup automated (daily)	☐

7.2 Launch Announcement Plan

- Write a dedicated "I built this" post explaining tech stack + journey
- Post on LinkedIn with screenshots and learnings
- Submit to r/netsec, r/webdev, and relevant Discord communities
- Reach out to 3–5 cybersecurity mentors or influencers for feedback

7.3 Post-Launch Iteration Backlog

Feature	Description	Priority
Newsletter	Email subscription with Resend or ConvertKit	High
Search	Full-text search across all posts (Algolia or Pagefind)	High
Comments	Giscus (GitHub Discussions) for blog comments	Medium
Vulnerability DB	Personal CVE notes and research database	Medium
Security Tools	Browser-based tools (port scanner, hash checker)	Medium
Dark/Light toggle	Persist preference in localStorage	Low
Reading time	Estimated read time on each post	Low
Certification wall	Display earned certs with verification links	Low

8. Learning Resources for NAMLAMEIS

8.1 Technical Skills to Build in Parallel

Skill	Resource	Level
TypeScript	TypeScript official docs + Matt Pocock exercises	Beginner→Intermediate
React / Next.js	official nextjs.org/learn course	Beginner→Intermediate
Web Security	PortSwigger Web Security Academy (free!)	All levels
Linux / CLI	overthewire.org Bandit wargame	Beginner
Networking	Professor Messer CompTIA Network+ (free videos)	Beginner
Git / GitHub	Pro Git book (free at git-scm.com/book)	Beginner→Intermediate
Python scripting	Automate the Boring Stuff with Python (free)	Beginner
CTF practice	TryHackMe.com learning paths	All levels

8.2 Recommended Certifications Roadmap

- CompTIA A+ — Foundation (optional for juniors, good for fundamentals)
- CompTIA Network+ — Networking essentials (6–12 months in)
- CompTIA Security+ — First major security cert, widely recognized
- eJPT (eLearnSecurity) — Practical beginner penetration testing
- CEH or OSCP — After 1–2 years of experience

9. Risk Register

Risk	Likelihood	Impact	Mitigation
Scope creep — adding features before core is done	High	Medium	Stick to phase gates; backlog all new ideas
Burnout from solo development	Medium	High	Time-box work; celebrate small wins
Security breach of the website itself	Low	High	Phase 3 hardening; regular audits
Writer's block — no content ideas	Medium	Medium	Maintain a backlog of 20+ topic ideas
Third-party dependency vulnerability	Medium	Medium	Dependabot + weekly npm audit
Poor SEO — no organic traffic	Medium	Low	Focus on technical SEO from Phase 4
Domain or hosting cost overrun	Low	Low	Free tier Vercel covers early-stage traffic

10. Success Metrics — End of Year 1

Metric	Target	Why It Matters
Blog posts published	≥ 20 posts	Establishes content credibility
Monthly organic visitors	500+ sessions/month	Proves SEO is working
Lighthouse avg score	≥ 90 on all pages	Technical quality signal
Security headers grade	A+ on SecurityHeaders.com	Demonstrates security knowledge
GitHub stars on tools repo	≥ 25 stars	Community validation
Newsletter subscribers	≥ 100 subscribers	Audience building metric
LinkedIn connections	≥ 300 relevant connections	Professional network growth

You got this, NAMLAMEIS. Build in public. Break things safely. Grow every day. 

Roadmap v1.0 — Prepared for NAMLAMEIS | Junior Developer Cybersecurity Track